

Experiment for the thesis "GENERATIVE AI FOR CLASS DIAGRAM MODELING FROM SOFTWARE REQUIREMENTS"

Dear Participant,

Thank you for taking part in this experiment, which is part of a MSc dissertation on the use of Generative AI (GAI) tools to derive the class diagrams from software requirements.

You will be asked to evaluate class diagrams modelled based on a three quality aspects (syntactic, semantic, and pragmatic), and rate each item in a scale **1–5**, where **1 = lowest** and **5 = highest**.

Syntactic Quality: assesses if the class diagrams follow the syntactic structure of UML Class Diagrams. (Missing cardinality details, and inappropriate naming of classes and associations are examples of rules verified for syntactic quality.)

Semantic Quality: evaluates the accuracy and completeness of the diagrams in representing the intended domain, by measuring (i) validity: all elements and relationships in the diagrams should accurately represent the domain; and (ii) completeness: the diagrams should include all necessary elements and relationships. (Incorrect cardinality, aggregation instead of association, wrong assignment of attributes or operations, operations cannot be realised using existing attributes and relationships, are examples of semantic issues.)

Pragmatic Quality: focuses on the understandability of the diagrams from the stakeholders' perspective. (Redundant attributes and associations, specialisation with no distinction among subclasses, inconsistency in styling and conventions are examples of pragmatic quality.)

To evaluate each class diagram, please select the model that, in your opinion, better models the requirements provided, and whether the model was AI generated or Human generated. There will be 3 different systems each with two class diagram, one created by Generative AI and another created by a human.

This task includes a brief explanation and example to guide your responses. This activity should take approximately 20 minutes to complete.

Your responses will be analyzed and compared with outputs generated by selected GAI tools (ChatGPT, Gemini, and Grok). The goal is to evaluate how accurately, completely, and effectively these tools perform compared to responses produced by humans, including both RE professionals and students with knowledge in the field.

Important: This experiment is designed to evaluate the performance of Generative AI tools, **not to assess or judge participants**. We are only interested in understanding how GAI-generated outputs compare to human-authored responses in terms of completeness, clarity, and relevance.

To ensure the validity of the experiment, **please do not use AI tools or assistants** to produce your answers. All responses should reflect your own knowledge, expertise, and reasoning.

Your participation is **anonymous** and all responses will be anonymized and analyzed in aggregate, strictly for academic purposes.

In case of doubt please contact tgb.rodriques@campus.fct.unl.pt

Thank you for your valuable contribution to this research!

Best regards,

Tiago Rodrigues

An AI-Powered Requirements Engineering Approach | FCT NOVA

Advisers: João Araújo and Ana Moreira, FCT NOVA

* Indica uma pergunta obrigatória

Participant information

1. What is your current role or title? *

(Example: Business Analyst, Requirements Engineer, Software Developer, Master's Student).

2. How many years of experience do you have in Requirements Engineering (RE)? *

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- ☐ Less than 1 year
- ☐ 1-3 years
- ☐ 4-6 years
- ☐ 7-10 years
- ☐ More than 10 years

3. What is your educational background? *

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- ☐ Bachelor's degree in Computer Science or related field
- ☐ Master's degree in Computer Science or related field
- ☐ PhD in a relevant field
- ☐ Outra: _____

4. In which industry or sector are you currently working or studying? *

Marcar tudo o que for aplicável.

- ☐ Software Development
- ☐ Academia / Research
- ☐ Finance
- ☐ Public Sector / Government
- ☐ Telecommunications
- ☐ E-commerce / Retail
- ☐ Student (Bachelor's or Master's)
- ☐ Outra: _____

Auto Repair System

The following class diagrams were created based on the system description provided next. Please read this description carefully and then evaluate the two class diagrams using the following criteria:

Syntactic Quality – Check if the class diagrams follow the syntactic rules of UML Class Diagrams. Look for: Missing cardinality details; Inappropriate naming of classes and associations

Semantic Quality – Check whether the diagrams accurately and completely represent the intended domain. Consider (i) *Validity* – Do all the elements and relationships accurately represent the domain? (ii) *Completeness* – Are all the necessary elements and relationships included? Look for: Incorrect cardinality; Aggregation used instead of association; Attributes or operations placed in the wrong location; Operations that cannot be realised with the given attributes and relationships

Pragmatic Quality – Check how understandable the diagrams are from the stakeholders' perspective. Look for: Redundant attributes and associations; Specialisations where subclasses are not clearly distinct; Inconsistencies in styling or conventions

System Description

Auto Repair

An auto repair shop, that sells and mounts parts and accessories for all kinds of vehicles, wants a new information system to manage their clients, parts, accessories and assembly services.

There are several employees. Each one of them has an unique identifying number, a name and an address.

In this shop, assembly services, where parts and accessories are installed in a vehicle, are executed. For each one these services the following data must be stored: In which car the service was executed, how many kms had the car at the time, who was the responsible employee, which parts and accessories were fitted, how many work hours did it take and the admission and finish dates.

Parts and accessories are only sold together with an assembly service.

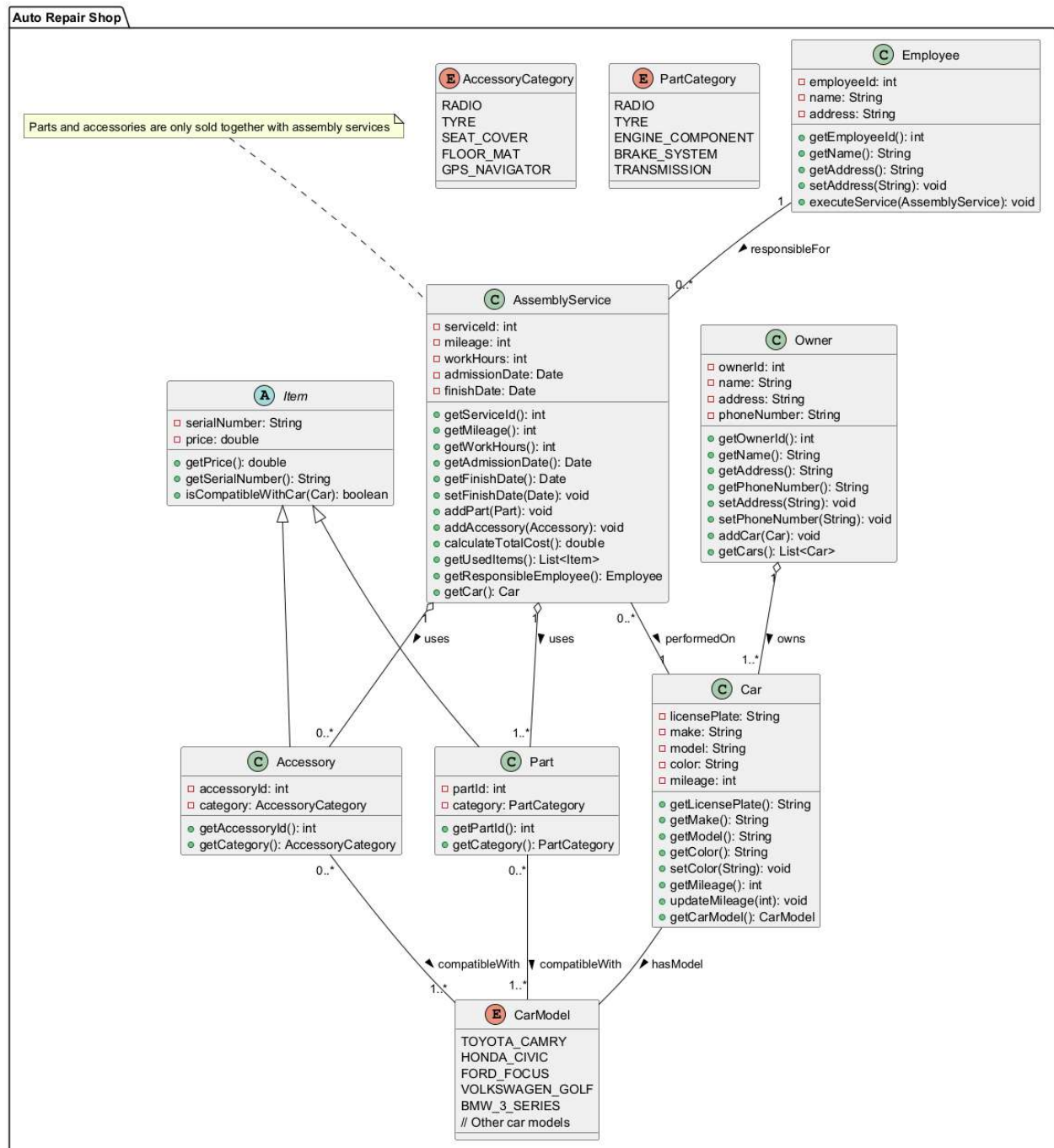
Each part/accessory only fits in some car models. Therefore, it is important to store that information.

Each part/accessory has a category (radio, tyre, ...), a serial number and a price.

Each car has a license plate, a make, a model, a color and an owner.
Each owner has a name, identifying number, address and a phone.

One person can own more than one car but one car only has one owner.

First Class diagram



Note

If the image is hard to read, open the full-size version to zoom or download. Here is the [link](#) for the full size version

5. Syntatic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

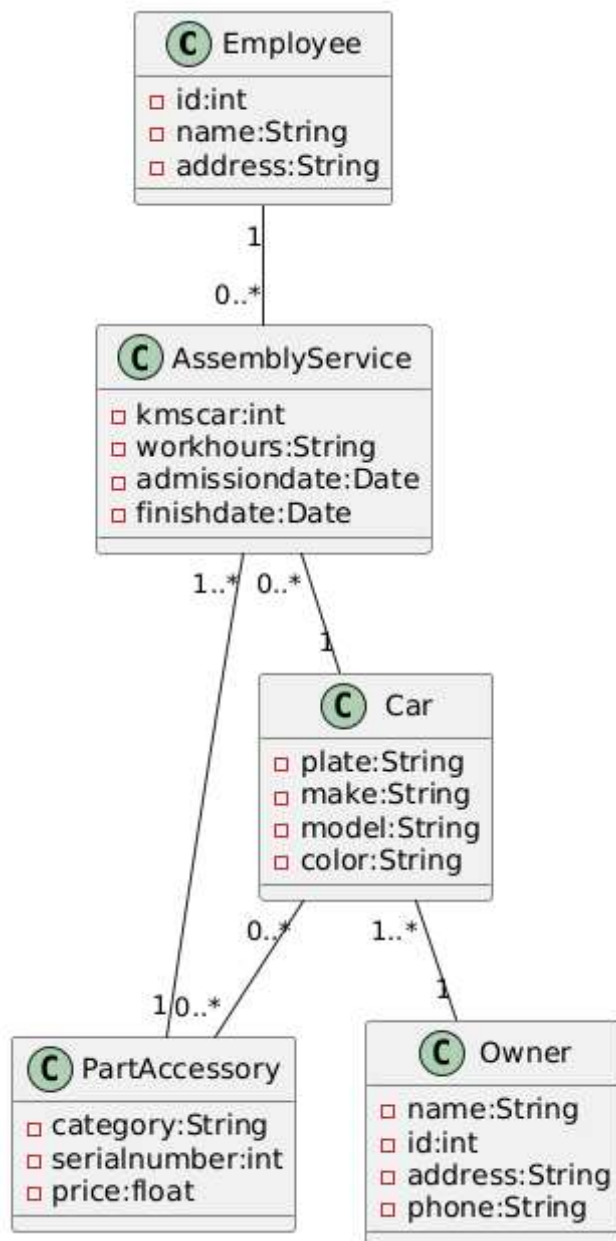
6. Semantic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

7. Pragmatic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

Second Class diagram



Note

If the image is hard to read, open the full-size version to zoom or download. Here is the [link](#) for the full size version

8. Syntatic Quality *

1 2 3 4 5



9. Semantic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

10. Pragmatic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

11. Which of the class diagram was created with Generative AI tools ? *

Marcar apenas uma oval.

- ☐ First Class diagram
- ☐ Second Class diagram

12. Select the one that best represents the system description. *

Marcar apenas uma oval.

- ☐ First Class diagram
- ☐ Second Class diagram

OOBank

The following class diagrams were created based on the system description provided next. Please read this description carefully and then evaluate the two class diagrams using the following criteria:

Syntactic Quality – Check if the class diagrams follow the syntactic rules of UML Class Diagrams. Look for: Missing cardinality details; Inappropriate naming of classes and associations

Semantic Quality – Check whether the diagrams accurately and completely represent the intended domain. Consider (i) *Validity* – Do all the elements and relationships accurately represent the domain? (ii) *Completeness* – Are all the necessary elements and relationships included? Look for: Incorrect cardinality; Aggregation used instead of association; Attributes or operations placed in the wrong location; Operations that cannot be realised with the given attributes and relationships

Pragmatic Quality – Check how understandable the diagrams are from the stakeholders' perspective. Look for: Redundant attributes and associations; Specialisations where subclasses are not clearly distinct; Inconsistencies in styling or conventions

System Description

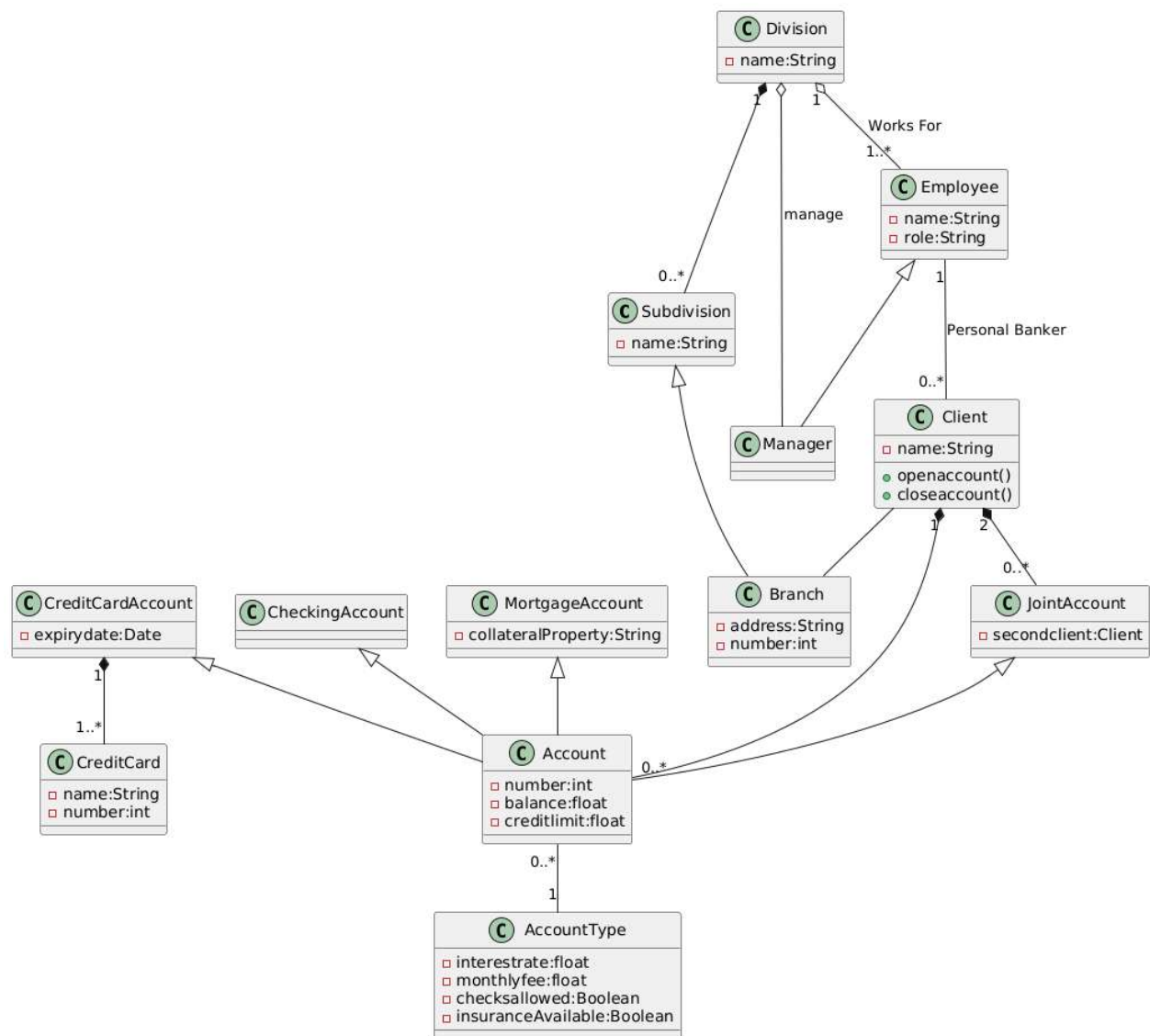
This system provides the basic services to manage bank accounts at a bank called OOBank. OOBank has many branches, each of which has an address and branch number.

A client opens accounts at a branch. Each account is uniquely identified by an account number; it has a balance and a credit or overdraft limit. There are many types of accounts, including: a mortgage account (which has a property as collateral), a checking account, and a credit card account (which has an expiry date and can have secondary cards attached to it).

It is possible to have a joint account (e.g. for a husband and wife). Each type of account has a particular interest rate, a monthly fee and a specific set of privileges (e.g. ability to write checks, insurance for purchases etc.).

OOBank is divided into divisions and subdivisions (such as Planning, Investments and Consumer); the branches are considered subdivisions of the Consumer Division. Each division has a manager and a set of other employees. Each customer is assigned a particular employee as his or her 'personal banker'.

First Class diagram



Note

If the image is hard to read, open the full-size version to zoom or download. Here is the [link](#) for the full size version

13. Syntatic Quality *

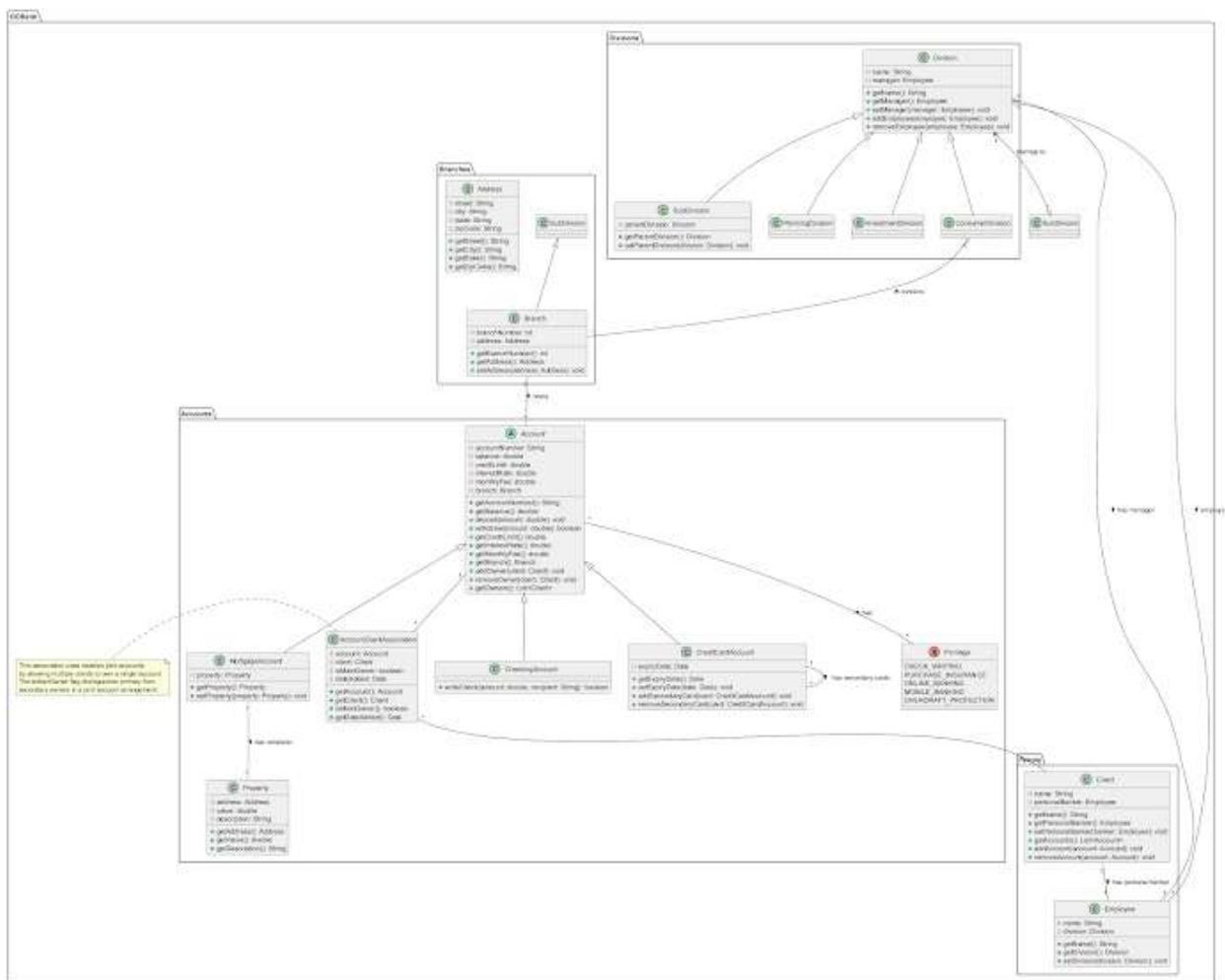
1	2	3	4	5
☆	☆	☆	☆	☆

1 2 3 4 5

☆☆☆☆☆

1 2 3 4 5

☆ ☆ ☆ ☆ ☆



Note

If the image is hard to read, open the full-size version to zoom or download. Here is the [link](#) for the full size version

16. Syntatic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

17. Semantic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

18. Pragmatic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

19. Which of the class diagrams was created with Generative AI tools? *

Marcar apenas uma oval.

- ☐ First Class diagram
- ☐ Second Class diagram

20. Select the one that best represents the system description. *

Marcar apenas uma oval.

☐ First Class diagram

☐ Second Class diagram

Online Shopping

The following class diagrams were created based on the system description provided. Please read the system description carefully and then evaluate the two class diagrams using the criteria below:

Syntactic Quality – Check if the class diagrams follow the syntactic rules of UML Class Diagrams. Look for: Missing cardinality details; Inappropriate naming of classes and associations

Semantic Quality – Assess whether the diagrams accurately and completely represent the intended domain. Consider (i) *Validity* – Do all elements and relationships accurately represent the domain? (ii) *Completeness* – Are all necessary elements and relationships included?

When evaluating semantic quality, pay attention to: Incorrect cardinality; Aggregation used instead of association; Attributes or operations placed in the wrong location; Operations that cannot be realized with the given attributes and relationships

Pragmatic Quality – Focus on how understandable the diagrams are from the perspective of stakeholders. Look for: Redundant attributes and associations; Specializations where subclasses are not clearly distinct; Inconsistencies in styling or conventions

System Description

Each customer has unique id and is linked to

exactly one **account**. Account owns shopping cart and orders.

Customer could register as a web user to be able to buy items online. Customer is not required to be a web user because purchases could also be made by phone or by ordering from catalogues. Web user has login name which also serves as unique id. Web user could be in several states - new, active, temporary blocked, or banned, and be linked to a **shopping cart**.

Shopping cart belongs to account.

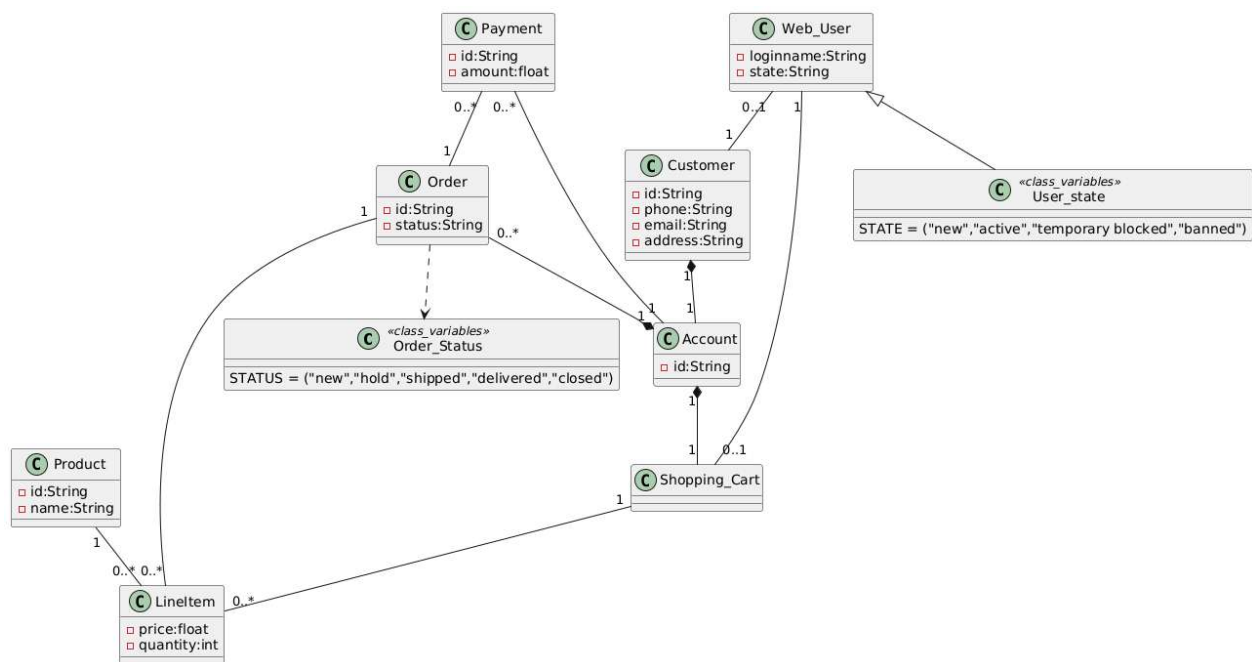
Account owns customer orders. Customer may have no orders.

Customer orders are sorted and unique. Each order could refer to several **payments**, possibly none. Every payment has unique id and is related to exactly one account.

Each order has current order status. Both order and shopping cart have **line**

items linked to a specific product. Each line item is related to exactly one product. A product could be associated to many line items or no item at all.

First Class diagram



Note

If the image is hard to read, open the full-size version to zoom or download. Here is the [link](https://docs.google.com/forms/d/1RenjrlceO8DdSYMrt_awaZx48Xa3RtqpNfrKWbf_Gm8/edit#settings) for the full size version

21. Syntatic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

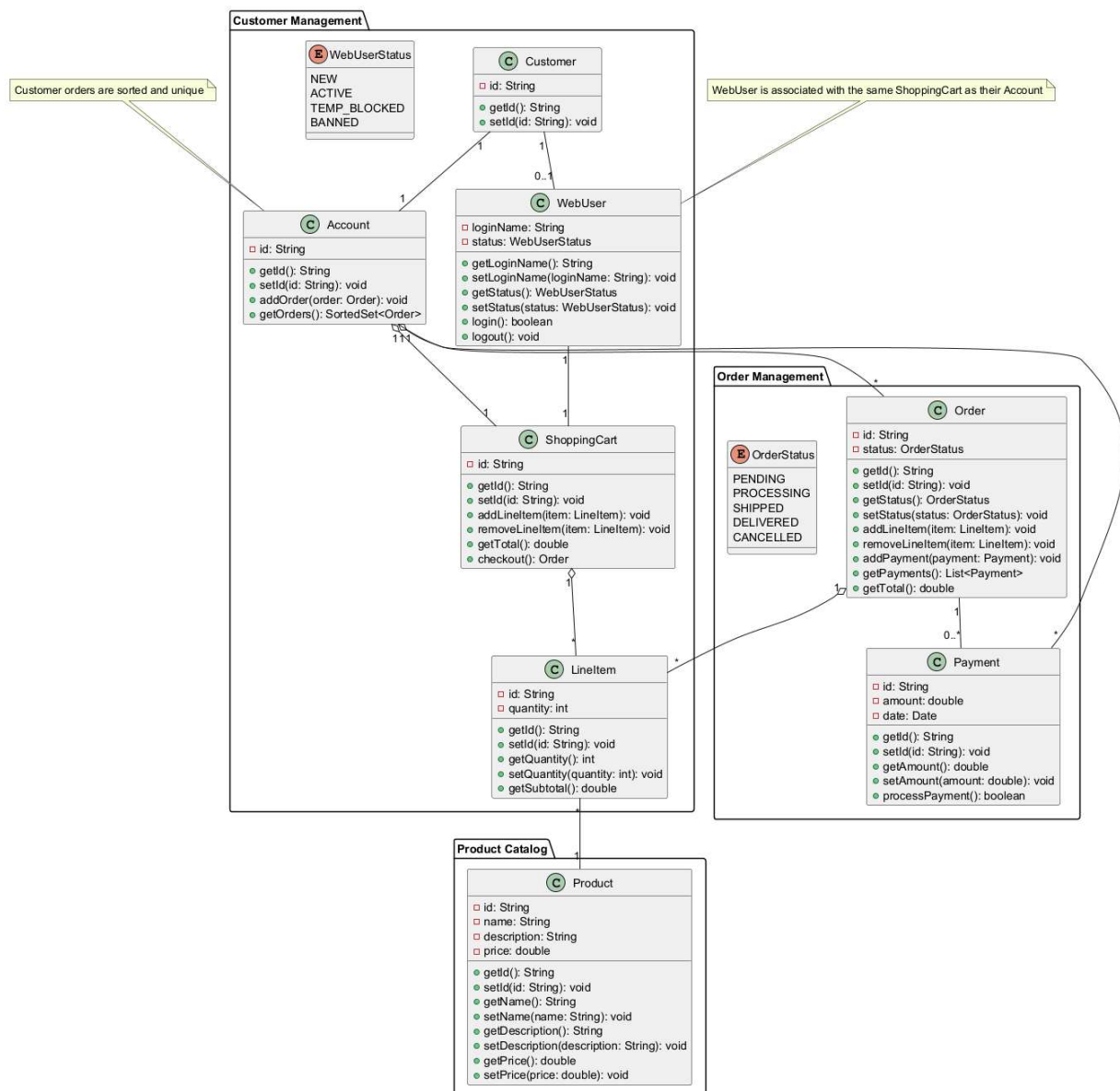
22. Semantic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

23. Pragmatic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

Second Class diagram



Note

If the image is hard to read, open the full-size version to zoom or download. Here is the [link](#) for the full size version

24. Syntactic Quality *

1 2 3 4 5



25. Semantic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

26. Pragmatic Quality *

1	2	3	4	5
☆	☆	☆	☆	☆

27. Which of the class diagram was created with Generative AI tools ? *

Marcar apenas uma oval.

- ☐ First Class diagram
- ☐ Second Class diagram

28. Select the one that best represents the system description. *

Marcar apenas uma oval.

- ☐ First Class diagram
- ☐ Second Class diagram

Consent and Participation Rules

29. Have you read the project description and the instructions for the task? *

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- ☐ Yes
- ☐ No

30. Do you confirm that you didn't use any AI-based tools (e.g., ChatGPT, Gemini, Bard, Copilot) to complete the task? *

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☐ Yes

☐ No

☐ Outra: _____

31. Do you agree to the use of your anonymized responses in aggregated format with other answers for academic research purposes only? *

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☐ Yes

☐ No

Thank you for your collaboration!

If you have any further questions regarding this form or the topic under study, please feel free to contact me at: **tgb.rodriques@campus.fct.unl.pt**

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